

COMPREHENSIVE_PROJECT_AUDIT_2026

HelixCode Comprehensive Project Audit Report 2026

Generated: 2026-01-10 **Audit Scope:** Full codebase analysis comparing documentation vs implementation **Status:** In Progress - Phase 7 of 8

Executive Summary

This audit analyzed **4,013 markdown files**, **73 SQL definitions**, **203 test files**, **42 internal packages**, and **514 Go source files** to identify gaps between documentation and implementation.

Key Metrics

Category	Status	Details
Overall Completion	~85%	Core features implemented
Test Coverage Average	68.7%	Requires improvement to 100%
Documentation Quality	95%	All packages have READMEs
Dependency Health	Excellent	All dependencies properly integrated
Critical Issues	12	Blocking production deployment
High Priority Issues	28	Should be fixed before release
Medium Priority Issues	45	Quality improvements
Low Priority Issues	30	Polish and completeness

SECTION 1: CRITICAL ISSUES (SHOWSTOPPERS)

1.1 Failing Tests

Status: BLOCKING

Unit Test Failure

Test	File	Error
TestExtractPythonSymbols	internal/repomap/ repomap_test.go	TempDir RemoveAll cleanup failure

Integration Tests Failing (Require Running Server)

Test Suite	Package	Issue
TestBasicIntegration	tests/e2e/phase2	Server not running on :8080
TestServerCapabilities	tests/e2e/phase2	Server not running on :8080
TestEnvironmentValidation	tests/e2e/phase2	Connection refused
TestRealServerIntegration	tests/e2e/phase2	Timeout
TestMemorySystemIntegration	tests/e2e/phase3	Server not running
TestConversationMemory	tests/e2e/phase3	Timeout
TestMemorySearchAndRetrieval	tests/e2e/phase3	Timeout
TestMemoryPersistence	tests/e2e/phase3	Timeout
TestMemoryAnalytics	tests/e2e/phase3	Timeout
TestMemoryPrivacyAndSecurity	tests/e2e/phase3	Timeout
TestConcurrentProjectOperations	tests/e2e/phase3	Timeout
TestMemoryOptimization	tests/e2e/phase3	Timeout
TestResourceCleanup	tests/e2e/phase3	Test timed out (10m)

Note: E2E/Integration tests require a running server. These are expected to fail in isolated test runs.

Impact: Unit test failure blocks CI/CD; Integration tests need server infrastructure

1.2 Placeholder Implementations Returning to Users

Status: CRITICAL

File	Lines	Issue
internal/cognee/ performance_optimizer.go	1187-1256	11 compression/traversal methods return nil
internal/config/ config.go	566-568, 1464-1483	UpdateConfigFromMap, SaveTemplate, LoadTemplate are stubs
internal/llm/ model_discovery.go	1136-1149	Hardcoded model alternatives instead of API queries
internal/llm/ usage_analytics.go	566, 581, 583	Hardcoded placeholder metrics
internal/cognee/ host_optimizer.go	7-18	Complete stub returning unchanged config
internal/tools/mapping/ treesitter.go	61-62, 267-269	Placeholder parser implementation

1.3 Missing API Endpoints (Documented but Not Implemented)

Status: CRITICAL - Feature gap

Endpoint Category	Count	Impact
MCP Server Management	6 endpoints	MCP not REST- accessible
Workflow State/History	5 endpoints	No workflow control via API
Notification Management	5 endpoints	Notifications not user- configurable

1.4 Low Test Coverage Packages (Below 50%)

Status: CRITICAL - Quality issue

Package	Coverage	Required
internal/llm	44.9%	100%

Package	Coverage	Required
internal/memory/providers	46.1%	100%
internal/workflow/planmode	39.7%	100%
shared/mobile-core	42.6%	100%
internal/tools/mapping	53.8%	100%
internal/providers	51.7%	100%

SECTION 2: HIGH PRIORITY ISSUES

2.1 Missing Feature Implementations

Feature	Documentation	Implementation Status
Advanced Reasoning API	CLI_Specs_5.md	NOT IMPLEMENTED
GenerateWithReasoning	Section 2, lines 71-82	No code exists
Chain-of-Thought	Specified	Not found
Tree-of-Thoughts	Specified	Not found

2.2 Documentation vs Code Inconsistencies

Missing CLI Commands (Documented in CLAUDE.md)

- `helix worker` command group (worker management)
- `helix workflow` command group (workflow control)
- `helix notify` command group (notification management)
- `helix session` command group (session management)

Missing Configuration Sections

- `reasoning`: configuration block
- `mcp.servers`: registration mechanism
- `Centralized notifications`: configuration

2.3 Silent Error Handling Issues

File	Lines	Issue
internal/hardware/detector.go	44-61	Returns success despite all detection failures

File	Lines	Issue
internal/task/ manager.go	257, 263, 302, 307	Returns nil, nil hiding cache errors
internal/tools/ multiedit/ transaction.go	354-369	Incomplete conflict verification

2.4 Applications Low Coverage

Application	Coverage	Required
aurora-os	4.9%	80%+
harmony-os	8.5%	80%+
desktop	9.1%	80%+
terminal-ui	9.6%	80%+
cmd/cli	11.9%	80%+

SECTION 3: MEDIUM PRIORITY ISSUES

3.1 Test Coverage Gaps (50-70%)

Package	Current	Target
internal/cognee	59.9%	100%
internal/focus	61.3%	100%
internal/mcp	61.3%	100%
internal/editor/formats	62.6%	100%
internal/tools/web	62.3%	100%
internal/workflow	64.7%	100%
internal/workflow/autonomy	65.0%	100%
internal/tools/confirmation	65.0%	100%
internal/tools/filesystem	66.8%	100%
internal/rules	66.8%	100%
internal/hardware	67.5%	100%
internal/memory	67.8%	100%
internal/llm/vision	68.3%	100%

Package	Current	Target
internal/tools/shell	68.2%	100%
internal/tools/multiedit	68.6%	100%
internal/agent/types	68.6%	100%
internal/redis	69.6%	100%

3.2 LLM Providers Without Dedicated Tests

Provider	File	Tests
Copilot	copilot_provider.go	NONE
KoboldAI	koboldai_provider.go	NONE
Llama.cpp	llamacpp_provider.go	NONE
Ollama	ollama_provider.go	NONE
OpenAI	openai_provider.go	NONE
OpenAI Compatible	openai_compatible_provider.go	NONE
OpenRouter	openrouter_provider.go	NONE
Tool Provider	tool_provider.go	NONE
xAI	xai_provider.go	NONE
Local Provider	local_provider.go	NONE

3.3 Workflow Templates With TODO Comments

Language	File	Lines	Issue
Go	executor.go	661-667	Generates TODO comment
Node.js	executor.go	696-700	Generates TODO comment
Python	executor.go	742-747	Generates TODO comment
Rust	executor.go	785-789	Generates TODO comment

SECTION 4: LOW PRIORITY ISSUES

4.1 Minor Version Conflicts

Dependency	Direct	Indirect	Issue
JWT	v4.5.2	v5.2.1	Dual version (functional)

Dependency	Direct	Indirect	Issue
Tree-sitter	0.0.0-2024...	-	Pre-release version

4.2 Packages with 0% Coverage (Utility/Scripts)

Package	Reason
cmd/config_test	Utility tool
cmd/helix_config	Management tool
cmd/performance_optimization	Optimization tool
cmd/security_fix	Security tool
examples/*	Example code
scripts/*	Build scripts
internal/mocks	Test infrastructure
internal/testutil	Test infrastructure

4.3 Documentation Improvements Needed

Package	Status	Action
internal/agent	No detailed README	Create documentation
internal/workflow	Basic README	Expand documentation
internal/session	Basic README	Expand documentation
internal/llm/vision	No README	Create documentation
internal/project	Basic README	Expand documentation

SECTION 5: TEST COVERAGE SUMMARY

Complete Coverage Data (from go test -cover ./...)

Packages at 100% Coverage (Target Achieved)

Package	Coverage
internal/llm/compressioniface	100.0%
internal/notification/testutil	100.0%
internal/provider	100.0%

Package	Coverage
internal/security	100.0%
internal/version	100.0%

Packages at 90%+ Coverage (Excellent)

Package	Coverage
internal/monitoring	97.1%
internal/agent/task	96.6%
internal/session	95.0%
internal/hooks	93.4%
internal/logo	92.6%
internal/template	92.1%
internal/context/mentions	91.4%
internal/fix	91.0%
internal/context/builder	90.0%

Packages at 80%+ Coverage (Good)

Package	Coverage
internal/performance	89.4%
internal/commands/builtin	88.9%
internal/discovery	88.4%
internal/editor	87.9%
internal/event	84.5%
internal/deployment	83.8%
internal/context	83.8%
internal/logging	83.3%
internal/auth	81.4%

Packages at 70-80% Coverage (Acceptable)

Package	Coverage
internal/notification	79.4%

Package	Coverage
internal/commands	78.1%
internal/persistence	77.7%
internal/project	77.0%
internal/llm/compression	75.2%
internal/database	73.8%
internal/worker	73.7%
internal/task	73.5%
internal/config	72.4%
internal/agent	72.2%
internal/repomap	71.2%
internal/workflow/snapshots	70.5%
internal/tools/git	70.5%

Packages at 60-70% Coverage (Needs Improvement)

Package	Coverage
internal/redis	69.6%
internal/tools/multiedit	68.6%
internal/agent/types	68.6%
internal/llm/vision	68.3%
internal/tools/shell	68.2%
internal/memory	67.8%
internal/hardware	67.5%
internal/tools/filesystem	66.8%
internal/rules	66.8%
internal/tools/confirmation	65.0%
internal/workflow/autonomy	65.0%
internal/workflow	64.7%
internal/editor/formats	62.6%
internal/tools/web	62.3%
internal/mcp	61.3%

Package	Coverage
internal/focus	61.3%
internal/tools/voice	60.2%

Critical: Packages Below 60% Coverage

Package	Coverage	Priority
internal/cognee	59.9%	High
internal/server	55.1%	High
internal/tools	55.5%	High
internal/tools/browser	55.1%	Medium
internal/tools/mapping	53.8%	High
internal/providers	51.7%	High
tests/e2e/challenges	49.0%	Medium
internal/memory/providers	46.1%	Critical
internal/llm	44.9%	Critical
shared/mobile-core	42.6%	Critical
internal/workflow/planmode	39.7%	Critical

Applications (Very Low - Needs Major Work)

Application	Coverage
applications/aurora_os	4.9%
applications/harmony_os	8.5%
applications/desktop	9.1%
applications/terminal_ui	9.6%
cmd/cli	11.9%
tests/e2e/phase3	16.2%

Packages with 0% Coverage (Intentional/Utility)

- cmd/, *examples*/, scripts/* - Entry points and utilities
 - internal/mocks, internal/testutil - Test infrastructure
-

SECTION 6: REMEDIATION PLAN

Phase 1: Critical Fixes (Week 1-2)

- ☐ Fix TestExtractPythonSymbols failing test
- ☐ Implement compression/traversal methods (cognee)
- ☐ Complete config watcher implementation
- ☐ Replace hardcoded model alternatives with API calls
- ☐ Fix silent error handling in hardware detector

Phase 2: Test Coverage Improvement (Week 3-6)

- ☐ Increase internal/llm coverage to 80%+
- ☐ Increase internal/memory/providers coverage to 80%+
- ☐ Increase internal/workflow/planmode coverage to 80%+
- ☐ Add tests for all 10 untested LLM providers
- ☐ Increase application test coverage to 50%+

Phase 3: API Completion (Week 7-10)

- ☐ Implement MCP server management endpoints
- ☐ Implement workflow state/history endpoints
- ☐ Implement notification management endpoints
- ☐ Add CLI commands for worker, workflow, notify, session

Phase 4: Feature Completion (Week 11-14)

- ☐ Implement Advanced Reasoning API

- ☐ Complete tree-sitter parser implementation
- ☐ Finish Zep provider query format
- ☐ Complete transaction verification

Phase 5: Documentation & Polish (Week 15-16)

- ☐ Create missing package READMEs
 - ☐ Update configuration documentation
 - ☐ Sync CLI documentation with commands
 - ☐ Create video course content
 - ☐ Update website documentation
-

SECTION 7: DEPENDENCY ANALYSIS

Integration Quality Summary

Dependency	Version	Integration	Status
Gin	v1.11.0	Excellent	Production Ready
Viper	v1.21.0	Excellent	Production Ready
PGX	v5.7.6	Very Good	Production Ready
Redis	v9.17.2	Excellent	Production Ready
JWT	v4.5.2	Good	Minor version conflict
WebSocket	v1.5.3	Excellent	Production Ready
Chromedp	v0.14.2	Very Good	Production Ready
Testify	v1.11.1	Excellent	Production Ready
Cobra	v1.8.0	Excellent	Production Ready
Fyne	v2.7.0	Excellent	Production Ready
Tview	v0.42.0	Excellent	Production Ready
Tree-sitter	0.0.0-...	Good	Pre-release version

SECTION 8: TRACKING CHECKPOINTS

Checkpoint System

This audit supports stop/resume capability. Each section can be verified independently.

Verification Checklist

Critical Issues Verification

- ☐ Run `go test ./internal/repomap/...` - must pass
- ☐ Review `cognee/performance_optimizer.go` implementations
- ☐ Verify config methods are functional
- ☐ Confirm no hardcoded data in LLM responses

Coverage Verification

- ☐ Run `go test -cover ./...` - all packages 80%+
- ☐ Run `go test -cover ./internal/llm/...` - must be 80%+
- ☐ Run `go test -cover ./applications/...` - must be 50%+

API Verification

- ☐ Test all MCP endpoints exist
- ☐ Test all workflow endpoints exist
- ☐ Test all notification endpoints exist

Documentation Verification

- ☐ All packages have README.md
- ☐

- ☐ CLAUDE.md matches implemented features
 - ☐ API documentation matches handlers
-

SECTION 9: STATISTICS

Codebase Size

- **Total Go Files:** 514
- **Total Test Files:** 203
- **Total Lines of Code:** ~150,000+
- **Internal Packages:** 42
- **Applications:** 6 (server, cli, terminal-ui, desktop, aurora-os, harmony-os)

Documentation Size

- **Markdown Files:** 4,013
- **SQL Definitions:** 73
- **Configuration Files:** 50+
- **Diagram Files:** 30+ (embedded mermaid)

Test Distribution

- **Unit Tests:** 157 files
 - **Integration Tests:** 8 files
 - **E2E Tests:** 17 files
 - **Challenge Tests:** 8 files
 - **Security Tests:** 5 files
 - **Regression Tests:** 2 files
-

SECTION 10: NEXT ACTIONS

Immediate (Today)

1. Review this audit report
2. Prioritize critical issues
3. Assign owners to each section
4. Set up tracking in project management tool

This Week

1. Fix failing test (TestExtractPythonSymbols)

2. Begin test coverage improvement sprint
3. Review all placeholder implementations
4. Create tickets for missing API endpoints

This Month

1. Achieve 70% average test coverage
2. Implement missing API endpoints
3. Complete CLI command additions
4. Fix all silent error handling issues

Report Version: 1.0 **Last Updated:** 2026-01-10 **Next Review:** Weekly during remediation